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2/28/26

CS 362

Project 2: Question 1

My System Stats:

Processor: 13th Gen Intel(R) Core (TM) i7-1355U, 1700 MHz, 10 Cores, 12 Logical Processors

OS Architecture: AMD 64, 64-bit operating system, x64-based processor

Preparation: Using Visual Studio 2022 and command prompt to run the SATI.C file. Running each number of processors 5 times and averaging the time into a data graph.

GRAPH:

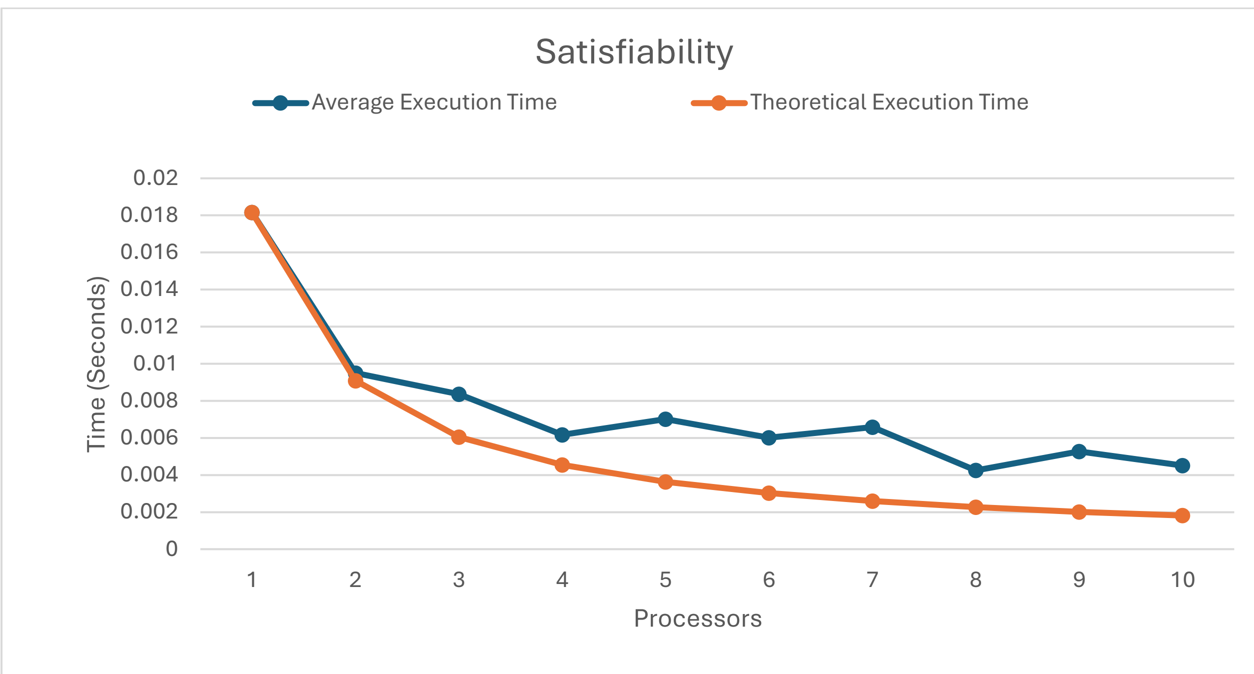

Table:

Number of Processor	Average Execution Time (Secs)	Theoretical Time (Secs)
1	0.0181588	0.0181588
2	0.0095048	0.0090794

3	0.0083594	0.0060529
4	0.00617	0.0045397
5	0.0070148	0.0036318
6	0.006013	0.0030265
7	0.006577	0.0025941
8	0.0042464	0.0022699
9	0.0052712	0.0020176
10	0.0045052	0.0018159

Summary:

Using the SATI.C file, the average amount of time it takes to process the file is about 0.01 seconds when using 2 processors. Based on the graph, the average processing time stays roughly between 0.01 and 0.006 seconds when using more than 2 processors. This suggests that the program performs best when using around 2–4 processors. While theoretically adding more processors should decrease processing time because the workload is divided among more units, the results do not show a significant improvement beyond 2 processors. After averaging the processing time five times for each processor count, the data indicates that the time saved becomes very small after 2 processors. This is likely due to overhead costs such as communication, synchronization, and task distribution between processors, which reduce the overall efficiency. Because of this, using more processors in this program leads to diminishing returns, making fewer processors more practical and efficient.